REX REMIGIUS STEPHEN JOTHI

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Education

Northeastern University, *Boston, MA* 2025 - Expected 2027
Master of Science in Computer Science
Relevant Course Work : Programming Design Paradigm, Computer Systems

Sri Krishna College of Engineering and Technology, *Coimbatore, India* 2019 - 2023
Bachelor of Technology in Information Technology (GPA: 8.68/10.00)
Relevant Course Work: Object Oriented Programming, Computer Architecture, Data Structures and Algorithms, Machine Learning Techniques and Big Data Analytics

Skills

Programming Languages: C, C++, Python, Java
Compiler & Systems: LLVM, GCC, Intrinsics Programming, SIMD Optimization
Machine Learning & AI Frameworks: TensorFlow, PyTorch, ONNX, TFLite, NumPy
Development & Testing Tools: Git/GitHub, PyTest, LIT, Docker, VSCode, Jupyter

Work Experience

***Software Engineer*, MulticoreWare Inc** Jan 2023 - Jul 2025

- Designed and implemented custom compiler intrinsics for LLVM and GCC, enabling code reuse across projects and improving execution efficiency by 15% through advanced optimization techniques and memory management strategies.
- Developed and integrated a novel optimization algorithm into LLVM's code generation pipeline, achieving a 5% performance improvement in generated binaries through enhanced instruction scheduling and register allocation.
- Architected and optimized machine learning kernels (convolution, maxpool, normalization) for a custom AI accelerator, reducing inference latency by 63% for single-stream workloads and 67% for multi-stream processing through efficient memory access patterns and SIMD optimization.
- Benchmarked and profiled multiple ML models (CNNs, transformers) on the AI accelerator architecture, identifying performance bottlenecks and implementing targeted optimizations that improved throughput by 40%.
- Awarded Employee of the Month twice for designing high-performance ML kernels for the AI accelerator and successfully migrating an audio processing model from PyTorch to TensorFlow and TFLite with zero accuracy loss.

***Summer Intern*, Zoho Corporation** Apr 2022 - May 2022

- Developed a Java command-line application with optimized data structures and algorithms, achieving 25% runtime performance improvement through efficient memory management and algorithmic optimization.
- Solved 100+ algorithmic problems across data structures, OOP design patterns, and the Java Collections Framework, strengthening debugging skills and understanding of computational complexity.

Projects

Enhanced Address Sanitizer pass in LLVM for fault tolerance system **C, LLVM, C++, Shell, Make**

- Enhanced LLVM's Address Sanitizer with fault-tolerant execution by implementing **CIMA (Countering Illegal Memory Access)** algorithm and thereby, achieving **100%** runtime availability, 0% crash rate, and **30%** lower reallocation overhead through custom CFG-based instrumentation and adaptive memory resizing.
- Integrated 3 new LLVM passes and plugin layers, improving error reporting and achieving **100%** detection accuracy across 20+ verified test cases covering heap, stack, and global memory errors.

Heart Disease Risk Prediction Model **Python, Pandas, NumPy, Matplotlib, Scikit-learn**

- Achieved **90%+** prediction accuracy (**94% ROC-AUC**) on UCI Heart Disease dataset by implementing ensemble learning methods that outperformed baseline logistic regression by 6%
- Optimized model performance through **960+ hyperparameter combinations** via Grid Search with 10-fold cross-validation, achieving balanced precision (89%) and recall (91%).

BYTES: Food Ordering App **Adobe XD, Figma, Dart**

- Engineered a food ordering application prototype using geofencing and Adobe XD, documenting findings to fix the three biggest causes of crashes and improving user experience by reducing the wait time **less than 10mins** improvements.